QUANTITATIVE TRADING

Model Evaluation Consolidation & Candidate Selection

Pilot equity: Lloyds Banking Group (LLOY.L) | Market data: Yahoo Finance
Evaluation status: validation only; final test excluded; no backtesting performed

1. Executive decision

Do not designate a single production winner at this stage. Retain Ridge Regression as the primary general-purpose candidate, retain the LSTM as a horizon-specific challenger for five- and ten-session forecasts, and keep LightGBM as a benchmark challenger. Retain the zero-return and historical baselines as mandatory reference models. This is a provisional model-selection decision for further validation, not approval for trading or deployment.

The evidence does not establish a statistically significant MAE advantage for LSTM over Ridge or LightGBM after Holm correction. LSTM is materially better than MLP on absolute and squared error in the reported paired tests, but MLP is not a competitive leading candidate.

2. Scope and evaluation protocol

Pilot instrument	LLOY.L (Lloyds Banking Group, London Stock Exchange; GBP).
Development period	24 October 2016 – 20 October 2023.
Validation period	23 October 2023 – 22 October 2025.
Final Test period	23 October 2025 – 23 October 2026; excluded from model selection and all results below.
Targets	Forward returns for horizons of 1, 5 and 10 sessions: Close(t+h) / Open(t+1) − 1.
Features	One-, five-, ten- and twenty-session close returns; five-, ten- and twenty-session close-return volatility.
Evaluation	Expanding walk-forward validation, weekly retraining/evaluation schedule, matured targets only, minimum 500 eligible training observations.
Comparability	Seven models compared on common date/horizon observations: 1,505 observations in total, with 506, 502 and 497 observations at horizons 1, 5 and 10 respectively.

3. Consolidated validation metrics

MAE and RMSE are in return units, not percentages. Lower is better for both. Directional Accuracy (DA) follows the project's canonical convention: actual > 0 and predicted > 0 are treated as positive; zero is treated as non-positive. Spearman measures rank association. Negative R^2 indicates performance worse than predicting the evaluation-sample mean under the usual R^2 definition.

Horizon: 1 session

Model	MAE	RMSE	DA	Spearman	R^2
Zero Return	0.009757	0.013655	N/A	N/A	-0.006043
Historical Mean	0.009759	0.013659	0.498024	-0.097032	-0.006630
Ridge Regression	0.009777	0.013688	0.500000	0.040177	-0.010919
Historical Median	0.009793	0.013693	0.462451	-0.093345	-0.011557
LightGBM	0.009854	0.013752	0.529644	0.105869	-0.020374
LSTM	0.010275	0.014327	0.501976	-0.000846	-0.107497
MLP	0.016136	0.021704	0.533597	-0.001242	-1.541662

Horizon: 5 sessions

Model	MAE	RMSE	DA	Spearman	R^2
LSTM	0.025819	0.035738	0.623506	0.123709	-0.031823
Ridge Regression	0.026023	0.035465	0.593625	0.178884	-0.016138
LightGBM	0.026133	0.035502	0.545817	0.158544	-0.018253
Historical Mean	0.026355	0.035839	0.583665	-0.135498	-0.037668
Zero Return	0.026485	0.035934	N/A	N/A	-0.043174
Historical Median	0.026620	0.036050	0.425000	-0.126595	-0.049921

MLP	0.031753	0.042175	0.484064	0.012936	-0.437011

Horizon: 10 sessions

Model	MAE	RMSE	DA	Spearman	R^2
Ridge Regression	0.039124	0.051289	0.561368	0.123284	-0.057729
Historical Mean	0.039502	0.051818	0.623742	-0.174932	-0.079656
LSTM	0.039562	0.051252	0.599598	0.163942	-0.056216
Zero Return	0.039843	0.052158	N/A	N/A	-0.093884
LightGBM	0.039940	0.052301	0.488934	0.075672	-0.099903
Historical Median	0.040012	0.052286	0.475610	-0.150928	-0.099245
MLP	0.044880	0.058873	0.503018	0.041432	-0.393688

4. Paired statistical comparison

The reported paired HAC tests use overlapping multi-session targets, with lag choices of 5, 4 and 9 for horizons 1, 5 and 10 in the current comparison script. These choices should be documented and reviewed for consistency before final sign-off. Holm-adjusted results indicate that LSTM has lower absolute error than MLP at all three horizons. No Holm-significant MAE advantage over Ridge or LightGBM was found. Directional comparisons are more sensitive to metric conventions and the handling of zero predictions; they should not be used as the principal selection criterion until the comparison code and reported tables are fully reconciled.

A statistical difference in forecast error is not evidence of economic profitability. No transaction costs, execution assumptions, portfolio constraints or strategy backtest are included in this evaluation.

5. Candidate selection

- **Primary general-purpose candidate: Ridge Regression.** Best overall compromise across the three horizons: lowest MAE among the learned models at horizons 1 and 10, and close to the LSTM at horizon 5. It has the strongest Spearman correlation among the listed learned models at horizon 5; LightGBM is higher at horizon 1 and LSTM is higher at horizon 10. Keep as the default benchmark candidate, subject to further validation.
- **Horizon-specific challenger: LSTM.** Lowest MAE at horizon 5 and competitive MAE at horizon 10. It is clearly better than MLP on error metrics, but no statistically significant MAE superiority over Ridge or LightGBM is established. Retain as a challenger, not as the sole selected model.
- **Secondary challenger: LightGBM.** Competitive baseline with the strongest learned-model MAE at horizon 1 after Ridge, and the strongest horizon-1 DA among the reported non-baseline models. Its overall evidence does not establish consistent superiority.
- **Mandatory references: Zero Return, Historical Mean, Historical Median.** Keep these baselines in every future evaluation. Zero Return has the lowest MAE at horizon 1, demonstrating how small the differences are at that horizon. Historical Mean has the highest reported DA at horizon 10, illustrating that DA alone can favour a model with weak rank association or negative R^2.
- **Do not advance as a leading candidate: MLP.** It has substantially worse MAE and RMSE than the leading candidates at all horizons and is significantly worse than LSTM in the reported paired error tests.

6. Interpretation and limitations

- All reported R^2 values are negative. The models have not demonstrated improvement over the evaluation-sample-mean reference in squared-error terms; this warrants caution about predictive value.
- Differences between MAE values are small for the leading models, and paired tests do not establish LSTM superiority over Ridge or LightGBM after multiple-testing correction.
- DA is convention-dependent. The canonical function treats zero actual/predicted returns as non-positive. Alternative conditional DA definitions must be labelled separately and must not be mixed with the canonical metric.
- The LSTM metric discrepancy has been traced to metric implementation/convention: the canonical function exactly reproduces the original LSTM DA values (0.501976, 0.623506, 0.599598). The comparison artefact must use this same canonical function.
- Historical baseline DA can be high even when rank correlation and R^2 are poor. DA should not be used alone to select a forecasting model.

- This is one equity and one validation window. Findings cannot be generalised to all NYSE, Nasdaq or LSE equities without stock-specific evaluation.
- No final-test evaluation, trading simulation, backtesting or profitability analysis has been performed.

7. Recommended next actions

1. Make src.baseline_evaluation._calculate_metrics the single canonical implementation for all model reports, including the LSTM comparison; regenerate only affected metric/comparison artefacts, not model training.
2. Add a regression test asserting that DA for the saved LSTM predictions reproduces 0.501976, 0.623506 and 0.599598 (within a numeric tolerance), and tests covering zero actual and zero predicted returns.
3. Re-run the paired comparison artefact generation after metric harmonisation and verify all model/horizon rows, p-value adjustment family definitions, HAC lag choices and bootstrap settings.
4. Record Ridge as the provisional primary candidate; retain LSTM and LightGBM as challengers; retain all three baselines.
5. Before any strategy or deployment phase, define economic evaluation protocol, trading costs, execution assumptions, risk limits and a strictly separate final-test protocol. No such activity is authorised by this report.

8. Decision record

Decision	Provisional candidate selection; no production model approved.
Primary candidate	Ridge Regression
Challengers	LSTM; LightGBM
Mandatory baselines	Zero Return; Historical Mean; Historical Median
Deprioritised model	MLP
Final Test	Excluded
Backtesting	Not performed
Evidence status	Validation comparison completed; metric harmonisation and statistical-reporting checks remain before sign-off.

Language note: This report is written in British English. It distinguishes observed validation results from recommendations and does not claim profitability.